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Inventor(s)	Ryu; Young-Bae et al.

Integrated actuator assembly for disconnecter device and electronic limited slip differential device

Abstract

An integrated actuator assembly for a disconnecter device and an electronic limited slip differential (eLSD) device includes a driving motor, a screw shaft rotated by the driving motor, a screw nut screw-coupled to the screw shaft and moving linearly in an axial direction of the screw shaft by the rotation of the screw shaft, a first fork unit engaged with the screw nut to linearly move a first sleeve of a disconnecter device, a second fork unit engaged with the screw nut to linearly move a second sleeve of an eLSD device, and a fork selection member for controlling the screw nut to be selectively engaged with any one of the first fork unit and the second fork unit.

Inventors: Ryu; Young-Bae (Hwaseong-si, KR), Seo; Kang-Soo (Hwaseong-si, KR), Lim; Jin-Seung (Hwaseong-si, KR), Jun; Jae-Weon (Hwaseong-si, KR), Baek; Sang-Bum (Hwaseong-si, KR)

Applicant: HYUNDAI MOTOR COMPANY (Seoul, KR); KIA CORPORATION (Seoul, KR)

Family ID: 1000008291817

Assignee: HYUNDAI MOTOR COMPANY (Seoul, KR); KIA CORPORATION (Seoul, KR)

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Primary Examiner: Wu; Lori

Attorney, Agent or Firm: McDonnell Boehnen Hulbert & Berghoff LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Korean Patent Application No. 10-2024-0091703, filed on Jul. 11, 2024, which is incorporated herein by reference in its entirety.

BACKGROUND

Field of the Disclosure

(2) The present disclosure relates to an integrated actuator assembly for a disconnecter device and an electronic limited slip differential device, which may sequentially operate the disconnecter device and electrode limited slip differential device for a vehicle using one actuator.

Description of Related Art

(3) Since four-wheel drive electric vehicles experience drag loss in a non-driven wheel when two wheels are driven or during coasting traveling, a disconnecter device for mechanically blocking connection between the non-driven wheel and a motor and a reducer is provided.

(4) The disconnecter device allows an electric vehicle to selectively travel using four wheels and two wheels to reduce no-load drag loss.

(5) In addition, a separate electronic limited slip differential (eLSD) device is installed to improve the traveling performance of the vehicle and compensate for the loss of a driving force due to

wheel slip, etc. As the eLSD device is provided, handling performance and starting performance are improved.

(6) Both the disconnecter device and the eLSD device need to be provided in the vehicle, which acts as a factor that increases the production cost and weight of the vehicle.

(7) In particular, when the disconnecter device and the eLSD device are installed inside the vehicle, there is a problem that many spaces, such as a space in which the disconnecter device and the eLSD device are installed and a space in which an actuator for operating the disconnecter device and the eLSD device is installed, are required.

SUMMARY

(8) The present disclosure has been invented to solve the above problems and is directed to providing an integrated actuator assembly for a disconnecter device and an electronic limited slip differential (eLSD) device, which can minimize a space required for installation and operation by sequentially operating the disconnecter device and the eLSD device using one driving motor.

(9) An integrated actuator assembly for a disconnecter device and an electronic limited slip differential (eLSD) device according to the present disclosure for achieving the above object includes a driving motor, a screw shaft rotated by the driving motor, a screw nut screw-coupled to the screw shaft and linearly moving in an axial direction of the screw shaft by the rotation of the screw shaft, a first fork unit engaged with the screw nut to linearly move a first sleeve of the disconnecter device, a second fork unit engaged with the screw nut to linearly move a second sleeve of the eLSD device, and a fork selection member configured to control the screw nut to be selectively engaged with any one of the first fork unit and the second fork unit.

(10) The fork selection member includes a guide groove formed in a surface of the screw nut along a trajectory set on the screw nut at a predetermined depth, and a guide bolt which guides linear movement of the screw nut and into which an end portion is inserted into the guide groove so that the screw nut rotates along a linear movement displacement of the screw nut.

(11) The guide groove is formed with a disconnecter device manipulation section formed in the axial direction of the screw nut, an eLSD manipulation section disposed at an interval in a circumference direction of the screw nut from the disconnecter device manipulation section and formed in an axial direction of the screw nut, and a connection section connecting one end of the eLSD manipulation section from the other end of the disconnecter device manipulation section.

(12) An angle formed between the connection section and the disconnecter device manipulation section or the eLSD manipulation section is an obtuse angle.

(13) The screw nut is formed with a control finger extending in the radial circumference direction of the screw nut so that the control finger is selectively engaged with any one of the first fork unit and the second fork unit.

(14) The screw nut rotates in the circumference direction of the screw nut so that the control finger is selectively engaged with any one of the first fork unit and the second fork unit.

(15) The first fork unit includes a first lug engaged with the control finger, and a first fork formed integrally with the first lug and coupled by being fitted into a circumference of the first sleeve, and the second fork unit includes a second lug engaged with the control finger, and a second fork formed integrally with the second lug and coupled by being fitted into a circumference of the second sleeve.

(16) The control finger is selectively engaged with any one of the first lug and the second lug.

(17) The first lug and the second lug are disposed at a predetermined angle in the circumference direction of the screw nut.

(18) A manipulation groove to which the control finger is engaged is formed in each of the first lug and the second lug.

(19) A center of the first sleeve matches a center of the second sleeve.

(20) According to the integrated actuator assembly for a disconnecter device and an electronic limited slip differential (eLSD) device of the present disclosure having the above configuration, it

is possible to minimize the space required for installation and operation by sequentially operating the disconnecter device and the eLSD device using one driving motor.

(21) In addition, by operating the disconnecter device and the eLSD device using one driving motor, it is possible to reduce the number of parts, simplify the structure, and save the production cost.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1 is a perspective view of a disconnecter device-electronic limited slip differential (eLSD) device in which an integrated actuator assembly for a disconnecter device and an eLSD device according to the present disclosure is used.

(2) FIG. 2 is a perspective view showing a state in which the integrated actuator assembly for a disconnecter device and an eLSD device according to the present disclosure has been separated from the disconnecter device-eLSD device.

(3) FIG. 3 is a plan view showing a guide groove of a screw nut.

(4) FIG. 4 is a perspective view, side view, and state view of the guide groove showing the integrated actuator assembly for a disconnecter device and an eLSD device according to the present disclosure in a state in which a guide bolt is located at P1 of the guide groove.

(5) FIG. 5 is a perspective view, side view, and state view of the guide groove showing the integrated actuator assembly for a disconnecter device and an eLSD device according to the present disclosure in a state in which the guide bolt is located at P2 of the guide groove.

(6) FIG. 6 is a perspective view, side view, and state view of the guide groove showing the integrated actuator assembly for a disconnecter device and an eLSD device according to the present disclosure in a state in which the guide bolt is located at P3 of the guide groove.

(7) FIG. 7 is a perspective view, side view, and state view of the guide groove showing the integrated actuator assembly for a disconnecter device and an eLSD device according to the present disclosure in a state in which the guide bolt is located at P4 of the guide groove.

(8) FIG. 8 is a perspective view, side view, and state view of the guide groove showing the integrated actuator assembly for a disconnecter device and an eLSD device according to the present disclosure in a state in which the guide bolt is located at P5 of the guide groove.

DETAILED DESCRIPTION

(9) Hereinafter, an integrated actuator assembly for a disconnecter device and an electronic limited slip differential device according to the present disclosure will be described in detail with reference to the accompanying drawings.

(10) The integrated actuator assembly for a disconnecter device and an electronic limited slip differential (eLSD) device according to the present disclosure includes a driving motor 31, a screw shaft 32 rotated by the driving motor 31, a screw nut 33 screw-coupled to the screw shaft 32 and moving linearly in an axial direction of the screw shaft 32 by the rotation of the screw shaft 32, a first fork unit 35 engaged with the screw nut 33 to linearly move a first sleeve 15 of a disconnecter device 10, a second fork unit 36 engaged with the screw nut 33 to linearly move a second sleeve 25 of an eLSD device 20, and a fork selection member for controlling the screw nut 33 to be selectively engaged with any one of the first fork unit 35 and the second fork unit 36.

(11) The disconnecter device 10 and the eLSD device 20 are configured integrally by coupling a hub and multi-plate clutch of the eLSD device 20 to a side gear of the disconnecter device 10 and have the actuator assembly installed at an outer side thereof.

(12) Only one actuator assembly is installed, and the disconnecter device 10 and the eLSD device 20 are manipulated by the first fork unit 35 and the second fork unit 36 that are installed in the actuator assembly, respectively. And, the disconnecter device 10 and the eLSD device 20 are

operated sequentially.

(13) The driving motor **31** is fixedly installed at one side of a vehicle. The driving motor **31** rotates in one direction or an opposite direction according to a signal input from a controller (not shown) so that the disconnecter device **10** and the eLSD device **20** are engaged or disengaged.

(14) The screw shaft **32** has threads formed on an outer surface thereof and is rotated by the driving motor **31**.

(15) The screw nut **33** is screw-coupled to the screw shaft **32**. Since the screw nut **33** is screw-coupled to the screw shaft **32**, when the rotation of the screw nut **33** is restricted by a guide bolt **34** to be described below, the screw nut **33** moves linearly in the axial direction of the screw shaft **32**.

(16) A control finger **33b** is formed to protrude in a radial direction of the screw nut **33**. A selected one of the disconnecter device **10** and the eLSD device **20** is manipulated by the control finger **33b**.

(17) The first fork unit **35** is engaged with the screw nut **33** to linearly move the first sleeve **15** of the disconnecter device **10**, and the second fork unit **36** is engaged with the screw nut **33** to linearly move the second sleeve **25** of the eLSD device **20**.

(18) The disconnecter device **10** and the eLSD device **20** are disposed so that an axial center of the first sleeve **15** matches an axial center of the second sleeve **25**. As the axial center of the first sleeve **15** matches the axial center of the second sleeve **25**, a compression direction of the multi-plate clutch in the eLSD device **20** becomes the same as the axial direction of the disconnecter device **10**.

(19) Structures of the first fork unit **35** and the second fork unit **36** are as follows.

(20) The first fork unit **35** includes a first lug **35b** engaged with the control finger **33b**, a first fork **35c** coupled by being fitted into a perimeter of the first sleeve **15** of the disconnecter device **10**, and a first rod **35a** connecting the first lug **35b** to the first fork **35c**.

(21) The first lug **35b** is formed with a manipulation groove **35ba** that accommodates the control finger **33b** so that the first lug **35b** is engaged with the control finger **33b**. Since the manipulation groove **35ba** is open in a circumference direction of the screw nut **33**. Accordingly, the control finger **33b** may be located inside the manipulation groove **35ba** or separated from the manipulation groove **35ba** by rotation. When the control finger **33b** is located inside the manipulation groove **35ba**, the control finger **33b** is engaged with the first lug **35b** so that the screw nut **33** and the first fork unit **35** linearly move together. For example, FIG. 5 shows that the control finger **33b** of the screw nut **33** is located inside the manipulation groove **35ba**, and FIG. 6 shows that the control finger **33b** of the screw nut **33** is separated from the manipulation groove **35ba** by rotation of the screw nut **33**.

(22) Since the first fork **35c** is coupled by being fitted into a circumference of the first sleeve **15**, when the first fork **35c** moves linearly, the first sleeve **15** moves linearly to manipulate the disconnecter device **10**.

(23) In particular, the first sleeve **15** may be located outside a housing **11** in the disconnecter device **10**, thereby reducing a length of the disconnecter device **10**.

(24) The first rod **35a** is fastened to the first lug **35b** and the first fork **35c** so that the first lug **35b** and the first fork **35c** are integrated. The linear movement of the first fork unit **35** may be guided using the first rod **35a**.

(25) The second fork unit **36** also includes a second lug **36b** engaged with the control finger **33b**, a second fork **36c** that is formed integrally with the second lug **36b** and coupled by being fitted into a circumference of the second sleeve **25**, and a second rod **36a** integrally fastened to the second lug **36b** and the second fork **36c**.

(26) The second lug **36b** is also formed with a manipulation groove **36ba**. When the control finger **33b** is located in the manipulation groove **36ba**, the control finger **33b** is engaged with the second lug **36b** so that the screw nut **33** and the second fork unit **36** linearly move together.

(27) Since the second fork **36c** is coupled by being fitted into the circumference of the second sleeve **25**, when the second fork **36c** moves linearly, the eLSD device **20** is manipulated to be

engaged or disengaged.

(28) The second rod **36a** is integrally fastened to the second lug **36b** and the second fork to move together. The linear movement of the second fork unit **36** may be guided using the second rod **36a**.

(29) However, the first fork unit **35** and the second fork unit **36** are not engaged with the screw nut **33** at the same time but are selectively engaged with any one of the first fork unit **35** and the second fork unit **36**.

(30) To this end, a fork selection member for allowing the screw nut **33** to be selectively engaged with any one of the first fork unit **35** and the second fork unit **36** is included.

(31) The fork selection member includes a guide groove **33a** formed in a surface of the screw nut **33** along a trajectory set on the screw nut **33** at a predetermined depth, and the guide bolt **34** whose end portion is inserted into the guide groove **33a** so that the screw nut **33** rotates according to the linear movement displacement of the screw nut **33**.

(32) The guide groove **33a** is formed in the surface of the screw nut **33** at a predetermined depth. The guide groove **33a** is generally formed in a longitudinal direction of the screw nut **33**, and in a section in which the control finger **33b** moves between the first lug **35b** and the second lug **36b**, is formed substantially in the circumference direction of the screw nut **33**.

(33) The guide bolt **34** is installed in a state of being fixed to the vehicle. The end portion of the guide bolt **34** is inserted into the guide groove **33a**.

(34) Therefore, the guide bolt **34** guides the linear movement of the screw nut **33** and rotates the screw nut **33** according to the linear movement displacement of the screw nut **33**.

(35) The guide bolt **34** controls the screw nut **33** while moving relatively with respect to the screw nut **33**. Here, when the guide bolt **34** moves relatively with respect to the screw nut **33**, it means that, when the guide bolt **34** is located from P1 to P5 in FIG. 3, the guide bolt **34** does not move to the right side of the drawing, but actually, the screw nut **33** moves to the left side of the drawing in a fixed state of the guide bolt **34**.

(36) In FIG. 3, a trajectory of the guide bolt **34** moving relatively with respect to the guide groove **33a** is marked by single-dotted lines. In FIG. 3, although it is shown that the guide bolt **34** moves in the guide groove **33a**, actually, the screw nut **33** moves with respect to the fixed guide bolt **34**.

(37) The guide groove **33a** includes a disconnecter device manipulation section S_D formed in the longitudinal direction of the screw nut **33**, an eLSD manipulation section S_L disposed at intervals from the disconnecter device manipulation section S_D in the circumference direction of the screw nut **33**, and a connection section S_C that connects one end of the eLSD manipulation section S_L to the other end of the disconnecter device manipulation section S_D. Referring to FIG. 3, locations of start and end points of each section are shown as P1 to P5 (however, P3 is located in the middle of the connection section), and a state in which these are projected onto a straight line is shown. The guide bolt **34** moves relatively along the set trajectory, and at this time, the screw nut **33** linearly moves as much as a displacement projected onto the straight line.

(38) Since the disconnecter device manipulation section S_D and the eLSD manipulation section S_L are formed linearly, the screw nut **33** converts a rotational movement of the screw shaft **32** into a linear movement to linearly move the first fork unit **35** or the second fork unit **36** and manipulate the disconnecter device **10** or the eLSD device **20**.

(39) Meanwhile, the first lug **35b** and the second lug **36b** are disposed at a predetermined angle in the circumference direction of the screw nut **33** so that the control finger **33b** is engaged with any one of the first lug **35b** and the second lug **36b** by rotation. Since the control finger **33b** is engaged with any one of the first lug **35b** and the second lug **36b** by rotation, the first lug **35b** and the second lug **36b** are disposed at the predetermined angle in the circumference direction of the screw nut **33**, and thus the control finger **33b** is engaged with any one of the first lug **35b** and the second lug **36b** by rotation.

(40) To this end, the disconnecter device manipulation section S_D and the eLSD manipulation section S_L are disposed at an interval in the circumference direction of the screw nut **33** in the

guide groove **33a**.

(41) The connection section **S_C** forms an obtuse angle with the axial direction of the screw nut **33**. In FIG. 3, as an angle α between the connection section and the eLSD manipulation section **S_L** is shown as being an obtuse angle, the connection section **S_C** forms the obtuse angle with the axial direction of the screw nut **33**. Since the disconnecter device manipulation section **S_D** is parallel to the eLSD manipulation section **S_L**, an angle between the connection section and the disconnecter device manipulation section **S_D** is also an obtuse angle.

(42) Therefore, the guide bolt **34** may smoothly move when moving from the disconnecter device manipulation section **S_D** to the eLSD manipulation section **S_L** or conversely moving from the eLSD manipulation section **S_L** to the disconnecter device manipulation section **S_D**.

(43) When the screw nut **33** linearly moves in a state in which the guide bolt **34** is located in the disconnecter device manipulation section **S_D**, the screw nut **33** moves linearly together with the first fork unit **35** to manipulate the disconnecter device **10**. That is, when the guide bolt **34** moves from **P1** to **P2**, the first fork unit **35** also moves from **P1** to **P2** so that the disconnecter device **10** is fastened.

(44) The screw nut **33** also rotates in a state in which the guide bolt **34** is located in the connection section **S_C** so that the control finger **33b** moves between the first lug **35b** and the second lug **36b**. As the screw nut **33** rotates, the control finger **33b** rotates to be separated from the first lug **35b** and engaged with the second lug **36b** or conversely, to be separated from the second lug **36b** and engaged with the first lug **35b**.

(45) When the screw nut **33** linearly moves in a state in which the guide bolt **34** is located in the eLSD manipulation section **S_L**, the screw nut **33** moves linearly together with the second fork unit **36** to manipulate the eLSD device **20**.

(46) An operation of the integrated actuator assembly for a disconnecter device and an eLSD device according to the present disclosure having the above configuration is as follows.

(47) FIG. 4 shows a state in which both the disconnecter device and the eLSD device are disengaged.

(48) The control finger **33b** is in a state of being engaged with the first lug **35b**, and the guide bolt **34** is located at **P1** in the guide groove **33a**. In addition, both the first fork unit **35** and the second fork unit **36** are in a state of maximally moving backward.

(49) FIG. 5 shows that the disconnecter device **10** is engaged, and the eLSD device **20** is continuously disengaged.

(50) In the state of FIG. 4, when the driving motor **31** is operated by a controller, the screw nut **33** moves forward as the screw shaft **32** rotates. Since the rotation of the screw nut **33** is restricted by the guide bolt **34**, when the screw shaft **32** rotates, the screw nut **33** moves linearly. The guide bolt **34** linearly moves the screw nut **33** while moving relatively in the disconnecter device manipulation section **S_D**.

(51) When the screw nut **33** moves forward in a state in which the control finger **33b** has been engaged with the first lug **35b**, the first fork unit **35** also moves forward as much as the displacement of the screw nut **33**. When the screw nut **33** moves forward so that the guide bolt **34** is located from **P1** to **P2** in the screw nut **33**, the first fork unit **35** also moves forward from **P1** to **P2** as much as the screw nut **33** moves forward (see FIG. 5).

(52) As the first fork unit **35** moves forward and the first sleeve **15** coupled to the first fork **35c** moves forward, the disconnecter device **10** is in a fastened state.

(53) In this case, the second fork unit **36** is in a disengaged state because it maintains an initial location.

(54) FIG. 6 shows a state in which the control finger **33b** moves from the first fork unit **35** to the second fork unit **36**.

(55) When the controller continuously operates the driving motor **31**, the screw shaft **32** also rotates together. In this case, the screw nut **33** also rotates as the guide bolt **34** moves from **P2** to **P4** in the

guide groove **33a**.

(56) FIG. **6** shows a state in which the guide bolt **34** is located at **P3**, which is a middle point of the connection section **S_C** of the guide groove **33a**.

(57) When the screw nut **33** rotates, the control finger **33b** also rotates together to be separated from the first lug **35b** and rotate toward the second lug **36b**. Since the control finger **33b** is separated from the first lug **35b** and is in a state before engaged with the second lug **36b**, the first fork unit **35** and the second fork unit **36** do not move in a state in which the guide bolt **34** is located in the connection section **S_C**, and the disconnecter device **10** and the eLSD device **20** maintain their previous states.

(58) FIG. **7** shows that the screw nut **33** rotates so that the control finger **33b** is engaged with the second lug **36b**, and the guide bolt **34** is located at **P4**, which is a front end of the eLSD manipulation section **S_L**.

(59) FIG. **8** shows a state in which the screw nut **33** moves forward so that the eLSD device **20** is fastened. When the screw nut **33** moves forward in a state in which the control finger **33b** is engaged with the second lug **36b**, the guide bolt **34** moves relatively from **P4** to **P5** along the eLSD manipulation section **S_L**.

(60) When the screw nut **33** moves forward so that the guide bolt **34** is located from **P4** to **P5**, the second fork unit **36** also moves forward as much as the screw nut **33** moves forward, and the second fork **36c** also moves forward from **P4** to **P5** (see FIG. **8**).

(61) As the second fork unit **36** moves forward and the second sleeve **25** coupled to the second fork **36c** moves forward, the eLSD device **20** is in a fastened state.

(62) As described above, as the driving motor **31** rotates in one direction, the disconnecter device **10** is first engaged, and then the eLSD device **20** is in a fastened state.

(63) Therefore, the disconnecter device **10** and the eLSD device **20** may be sequentially fastened using one actuator assembly.

(64) Meanwhile, to disengage the disconnecter device **10** and the eLSD device **20**, the controller rotates the driving motor **31** in an opposite direction.

(65) When the driving motor **31** rotates in the opposite direction, the screw nut **33** moves backward, and the actuator assembly operates in a direction opposite to the direction when fastened.

(66) Therefore, the screw nut **33** moves backward in a state in which the control finger **33b** has been engaged with the second lug **36b** so that the eLSD device **20** is first disengaged. In a state in which the control finger **33b** is separated from the second lug **36b** and engaged with the first lug **35b**, the disconnecter device **10** is disengaged by the backward movement of the screw nut **33**.

(67) Even when the disconnecter device **10** and the eLSD device **20** are disengaged, the eLSD device **20** and the disconnecter device **10** may be disengaged sequentially.

Claims

1. An integrated actuator assembly for a disconnecter device and an electronic limited slip differential (eLSD) device, the integrated actuator assembly comprising: a driving motor; a screw shaft configured to be rotated by the driving motor; a screw nut coupled to the screw shaft and configured to linearly move in an axial direction of the screw shaft by rotation of the screw shaft; a first fork unit engaged with the screw nut and configured to linearly move a first sleeve of the disconnecter device; a second fork unit engaged with the screw nut and configured to linearly move a second sleeve of the eLSD device; and a fork selection member configured to control the screw nut to be engaged with any one of the first fork unit and the second fork unit.

2. The integrated actuator assembly of claim 1, wherein the fork selection member includes: a guide groove formed in a surface of the screw nut along a trajectory set on the screw nut at a predetermined depth; and a guide bolt configured to guide linear movement of the screw nut,

wherein an end portion of the guide bolt is inserted into the guide groove so that the screw nut rotates along a linear movement displacement of the screw nut.

3. The integrated actuator assembly of claim 2, wherein the guide groove is formed with: a disconnecter device manipulation section formed in the axial direction of the screw nut; an eLSD manipulation section disposed at an interval in a circumference direction of the screw nut from the disconnecter device manipulation section, and formed in an axial direction of the screw nut; and a connection section connecting one end of the eLSD manipulation section to an end of the disconnecter device manipulation section.

4. The integrated actuator assembly of claim 3, wherein an angle formed between the connection section and the disconnecter device manipulation section or the eLSD manipulation section is an obtuse angle.

5. The integrated actuator assembly of claim 1, wherein the screw nut is formed with a control finger extending in a radial direction of the screw nut, and the control finger is engaged with any one of the first fork unit and the second fork unit.

6. The integrated actuator assembly of claim 5, wherein the screw nut rotates in a circumference direction of the screw nut to engage the control finger with any one of the first fork unit and the second fork unit.

7. The integrated actuator assembly of claim 5, wherein the first fork unit includes: a first lug engaged with the control finger; and a first fork formed integrally with the first lug and coupled to a circumference of the first sleeve by being inserted into the circumference of the first sleeve; and wherein the second fork unit includes: a second lug engaged with the control finger; and a second fork formed integrally with the second lug and coupled to a circumference of the second sleeve by being inserted into the circumference of the second sleeve.

8. The integrated actuator assembly of claim 7, wherein the control finger is engaged with any one of the first lug and the second lug.

9. The integrated actuator assembly of claim 8, wherein the first lug and the second lug are disposed at a predetermined angle in a circumference direction of the screw nut.

10. The integrated actuator assembly of claim 7, wherein a manipulation groove to which the control finger is engaged is formed in each of the first lug and the second lug.

11. The integrated actuator assembly of claim 1, wherein a center of the first sleeve aligns with a center of the second sleeve.
